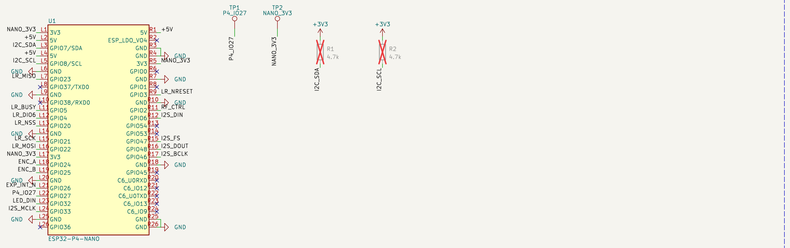
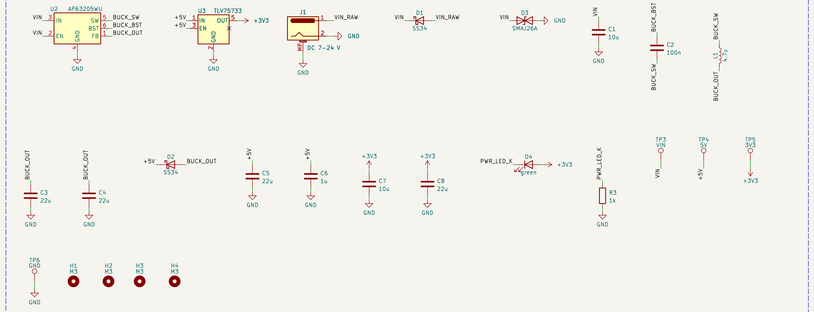
OpenTheatreComms base station — carrier PCB

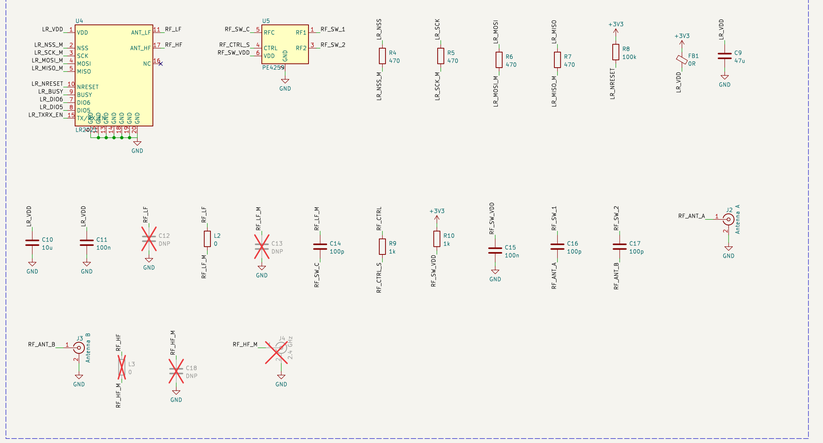
Waveshare ESP32-P4-NANO on two 2x13 sockets: Ethernet with a PoE module option, USB-C, Wi-Fi 6. Pins it uses itself are left open



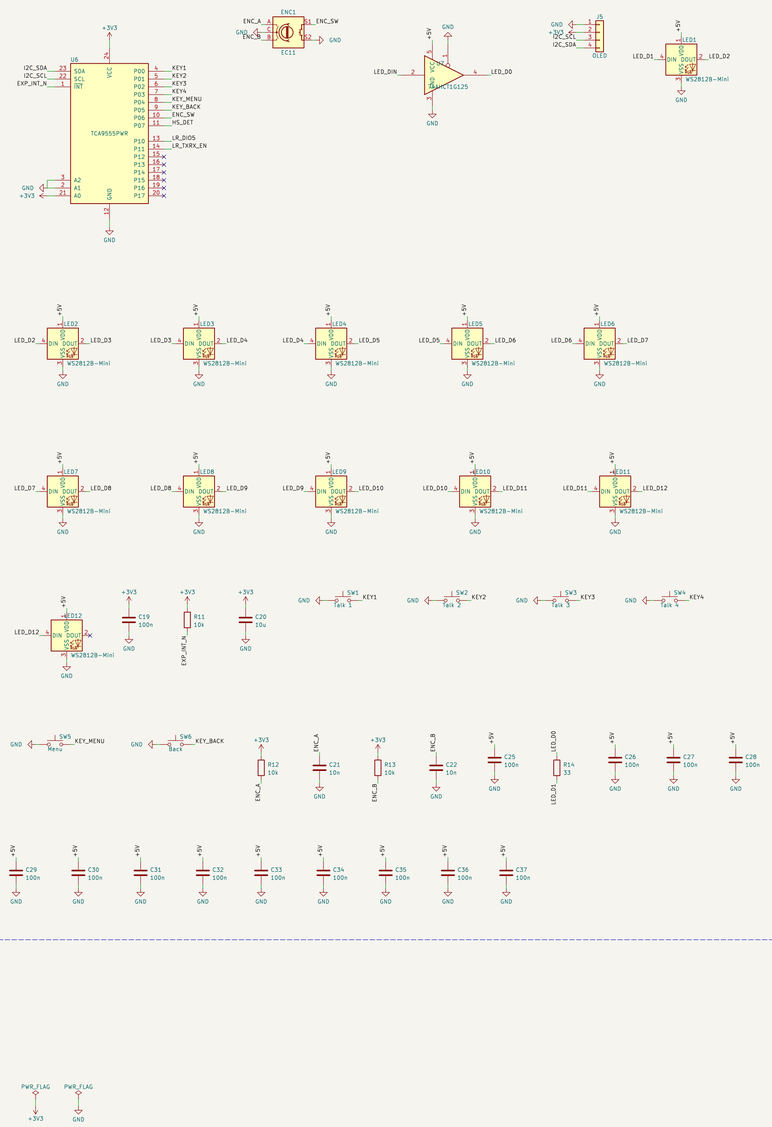
DC input (7–24 V) through a 5 V buck into the NANO's 5 V rail (USB-C or PoE also work), 3.3 V LDO for the carrier, mounting holes



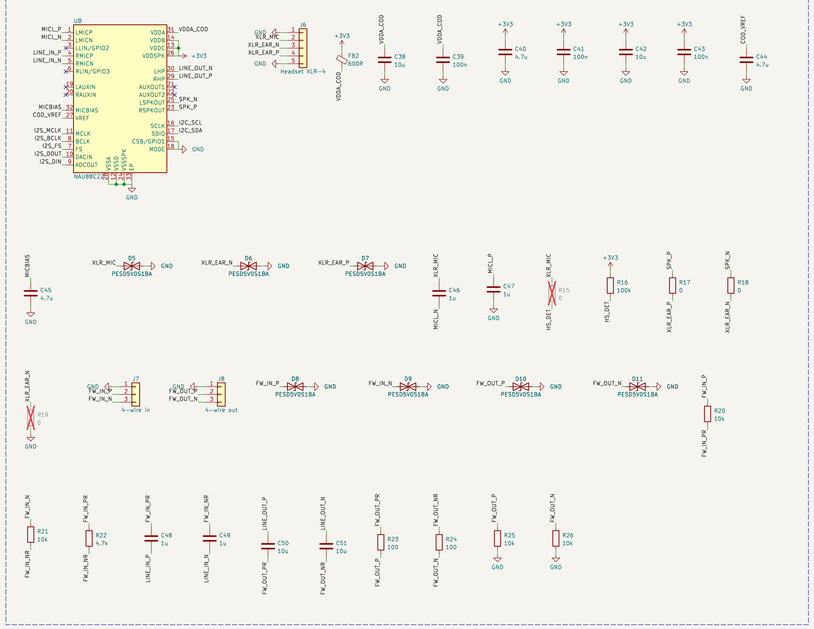
LR2021 module, 915 MHz pi match, PE4259 antenna diversity switch, two rear SMA jacks; 2.4 GHz port as a footprint only



IO expander (0x21), OLED socket, menu dial, four talk keys, Menu and Back keys, 12 RGB status lights on one data line



Optional NAU88C22 codec: front XLR-4 headset (BTL earpiece), rear balanced 4-wire line in and out



ESP32-P4-NANO carrier; power, LR2021 radio and RF, front panel, optional codec. Generated from pc

OpenTheatreComms

Sheet: /

File: base_pcb.kicad_sch

Title: Base station carrier PCB

Size: A1	Date: 2026-10-30	**Rev: 0.1**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1